Optimal Reconfiguration of Systems in Aeronautics

Lecture 02: “Handling qualities”

Bogdan D. Ciubotaru

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Outline

1. Handling Qualities for the Longitudinal Motion
2. Numerical Model for the Boeing 747 Aicraft

Objective I

This lecture provides the main lines for the characterization of the flight quality in terms of handling criteria.

Outline

1. Handling Qualities for the Longitudinal Motion
   - Quantitative Criteria for the Longitudinal Modes
2. Numerical Model for the Boeing 747 Aicraft

Performance quantification I

There are the so-called airworthiness parameters which provide the quantitative and qualitative description of the quality of flight. Literally, the ride quality parameters may be interpreted as measures of the effort done by the pilot to control the aircraft.

These parameters, precisely the control power and forces, and the static and dynamic stability, provide different levels of evaluation as functions of the complexity of particular flight phases, listed as follows:

- Category A: rapid maneuvering with precision tracking;
- Category B: gradual maneuvering without precision tracking;
- Category C: gradual maneuvering with precision flight-path control.

Still, since the pilot gives his personal evaluation, the most accepted universal scale providing the unifying framework for handling criteria analysis is the Cooper-Harper scale; let its quantitative result be denoted by CH.

In this respect, there are different levels of the quality of flight, listed in below:

- Level 1 - satisfactory: $\mathrm{CH} < 3.5$;
- Level 2 - acceptable: $3.5 < \mathrm{CH} < 6.5$;
- Level 3 - poor (but still controllable): $6.5 < \mathrm{CH} < 9.5(+)$.

Performance quantification II

In the sequel, short descriptions of parameters characterizing the quality of flight are given as follows:

i. Static Stability: it requires that the neutral point lie some distance behind the most aft position of the CoG. (The neutral point defines the location of the CoG at the boundary between its stable and unstable positions.)

ii. Dynamic Stability: it requires that the damping coefficient and undamped natural frequency characterizing the natural modes of evolution of the aircraft, respectively the short-period and phugoid for the longitudinal motion, take particular values in given sets. (For the lateral / directional motion, precisely for its natural modes of evolution, namely spiral, roll, and dutch-roll, other bounds are indicated for their dynamical parameters.)

However, other requirements may be imposed at the design stage, either on the limits of the control power or of the control forces; hence, for the control power, it amounts to defining the specific speed ranges that must be achievable with full elevator deflection, while, for the control forces, it demands either to specify the limits to which they must be exerted in order to effect specific changes from a given trimmed condition or to maintain the trim speed following a sudden change in configuration or throttle setting.

The standards for the flight quality in civil aviation are defined as FAR's (Federal Aviation Requirements) in US (precisely, FAR 23 and FAR 25), respectively JAR's (Joint Aviation Regulations) in EU.

Quantitative criteria for the short-period mode I

Recall the short-period system and control matrices and write the short-period mode characteristic equation

$$s^2 + 2\zeta_{SP}\omega_{SP}s + \omega_{SP}^2 = 0\,,$$

whose roots provide its dynamical parameters, namely the damping coefficient, ζ_{SP}, and the undamped natural frequency, ω_{SP}, as follows

$$2\zeta_{SP}\omega_{SP} = -\left(\frac{Z_w}{m} + \frac{M_q}{I_y} + \frac{M_{\dot{w}}u_0}{I_y}\right)\,,$$

$$\omega_{SP}^2 = \frac{Z_w M_q}{m I_y} - \frac{u_0 M_w}{I_y}\,,$$

and more

$$\zeta_{SP} = -\frac{1}{2}\left(\frac{Z_w}{m} + \frac{M_q}{I_y} + \frac{M_{\dot{w}}u_0}{I_y}\right)\sqrt{\frac{1}{\frac{Z_w M_q}{m I_y} - \frac{u_0 M_w}{I_y}}}\,,$$

$$\omega_{SP} = \sqrt{\frac{Z_w M_q}{m I_y} - \frac{u_0 M_w}{I_y}}\,,$$

Quantitative criteria for the short-period mode II

encountering the coarse approximation

$$\begin{aligned} 2\zeta_{SP}\omega_{SP} &\approx -\frac{M_q}{I_y}\,, \\ \omega_{SP}^2 &\approx -\frac{u_0 M_w}{I_y}\,, \end{aligned} \quad \Rightarrow \quad \begin{aligned} \zeta_{SP} &\approx -\frac{M_q}{2}\sqrt{-\frac{1}{u_0 M_w I_y}}\,, \\ \omega_{SP} &\approx \sqrt{-\frac{u_0 M_w}{I_y}}\,. \end{aligned}$$

Then, note that the damping coefficient ζ_{SP} depends primarily on M_q and the undamped natural frequency ω_{SP} on M_w (the measure of the aerodynamic stiffness in pitch), that is ω_{SP} will be higher for airplanes with a low pitching moment of inertia and, at any given altitude, it will be higher at high speed than at low speed.

Sofar, the short-period mode behavior is mainly evaluated based on the damping coefficient ζ_{SP}, which is indicated at $[0.2, 2]$ for Level 2 quality, and $\omega_{SP} \in [0.3, 2]$ for Level 1, respectively.

However, function of the allowable regions of the undamped natural frequency ω_{SP}, there are cases when limitations to the so-called Control Anticipation Parameter (CAP) are also indicated, where the latter is computed as follows

$$CAP = \frac{\omega_{SP}^2}{n_\alpha}\,,$$

for n_α being the horizontal load factor.

Quantitative criteria for the phugoid mode I

Now turn to the phugoid system and control matrices and write the phugoid mode characteristic equation

$$s^2 + 2\zeta_{PH}\omega_{PH}s + \omega_{PH}^2 = 0\,,$$

whose roots provide the dynamical parameters, namely the damping coefficient, ζ_{PH}, and the undamped natural frequency, ω_{PH}, as follows

$$2\zeta_{PH}\omega_{PH} = -\left[\frac{X_u}{m} + \frac{X_w}{m}\left(\frac{mu_0 M_u - Z_u M_q}{Z_w M_q - mu_0 M_w}\right)\right]\,,$$

$$\omega_{PH}^2 = g\left(\frac{Z_u M_w - Z_w M_u}{Z_w M_q - mu_0 M_w}\right)\,,$$

and more

$$\zeta_{PH} = -\frac{1}{2}\left[\frac{X_u}{m} + \frac{X_w}{m}\left(\frac{mu_0 M_u - Z_u M_q}{Z_w M_q - mu_0 M_w}\right)\right]\sqrt{\frac{1}{g\left(\frac{Z_u M_w - Z_w M_u}{Z_w M_q - mu_0 M_w}\right)}}\,,$$

$$\omega_{PH} = \sqrt{g\left(\frac{Z_u M_w - Z_w M_u}{Z_w M_q - mu_0 M_w}\right)}\,,$$

Quantitative criteria for the phugoid mode II

encountering the coarse approximation

$$\begin{aligned} 2\zeta_{PH}\omega_{PH} &\approx -\frac{X_u}{m}, \\ \omega_{PH}^2 &\approx -\frac{gZ_u}{mu_0}, \end{aligned} \quad \Rightarrow \quad \begin{aligned} \zeta_{PH} &\approx -\frac{X_u}{2m}\sqrt{-\frac{mu_0}{gZ_u}}, \\ \omega_{PH} &\approx \sqrt{-\frac{gZ_u}{mu_0}}. \end{aligned}$$

Then, the damping coefficient ζ_{PH} and the undamped natural frequency ω_{PH} express principal dependency on X_u and Z_u, respectively; note that the undamped natural frequency ω_{PH} depends only on the steady-state of speed and it is however independent of the design of the airplane; the damping coefficient ζ_{SP} is inversely proportional to the airplane so-called lift-to-drag (L/D) ratio and hence airplanes with high (L/D)s can be expected to have poor damping, which makes the control of speed difficult on final approach.

Sofar, the phugoid mode behavior is mainly evaluated based on the damping coefficient ζ_{PH}, which must satisfy $\zeta_{PH} > 0$ for Level 2, and $\zeta_{PH} > 0.04$ for Level 1 quality, respectively.

Moreover, the quality of the phugoid mode is said to be at Level 3 if the time-to-double $T_{2_{PH}}$ obeys $T_{2_{PH}} > 55\ sec$, where

$$T_{2_{PH}} = \frac{\ln 2}{-\zeta_{PH}\omega_{PH}}$$

represents the interval within which the phugoid mode will double its amplitude after some perturbation occurred, and $\ln(.)$ denotes the natural logarithm.

Remarks on handling criteria I

1. The flight quality quantifications are given for subsonic speeds, when the stability of the flight-path is almost equivalent to the stability in pitch; however, at supersonic flight these criteria change to the more appropriate pitch or flight-path bandwidth evaluations.
2. Still important in the analysis of stability and the quality of flight are the so-called stability slopes, given by the partial derivatives of the stability coefficients w.r.t. the principal angular variations.
3. The quantitative criteria are given in open-loop. Still, if an airplane reacts too slow or too fast to a pilot command, the pilot must compensate for this situation. Moreover, since he is influenced by the on-board indicators and acts consequently, his behavior may be interpreted as that of an intermediate servo-control system. Thus, there are attempts to model the human-factor as a linear, eventually optimal, system on the feed-forward path. In this perspective, the integration of the "pilot-in-the-loop" and the analysis of the "pilot-induced-oscillations" (PIO's) in the context of "man / machine" systems remain important matters in the area.
4. There are also several trials on implementing the handling criteria in software developed libraries of general use; however, their software integration in the so-called Total-In-Flight-Simulators (TIFS) rests challenging.

Outline

1. Handling Qualities for the Longitudinal Motion
2. Numerical Model for the Boeing 747 Aicraft
   - Geometrical and Aerodynamic Characteristics
   - Full Longitudinal Numerical Model
   - Short-Period Mode Numerical Model

Context I

In the following, particular values of physical data are provided in order to obtain the numerical model for the longitudinal motion of the Boeing 747 aircraft.

Geometrical and aerodynamic data I

First, the geometrical and aerodynamic data are given in Tab..

m	Aircraft total mass	19792	$slug$
W	Vehicle weight	636.636	lb
$\bar{c}$	Mean aerodynamic chord	27.31	ft
I_y	y moment of inertia	0.331×10^8	$slug \cdot ft^2$
u_0	Reference flight speed	774	fps
θ_0	Reference pitch angle	0	rad
g	Acceleration due to gravity	32.2	ft/sec^2
ρ	Density of air	0.0005909	$slug/ft^3$
h	Altitude relative to mean sea level	40000	ft

Aerodynamic and control derivatives I

Next, in Tab., the aerodynamic and control derivatives are provided.

	$X\,(lb)$	$Z\,(lb)$	$M\,(ft \cdot lb)$
$u\,(ft/sec)$	-1.358×10^2	-1.778×10^3	3.581×10^3
$w\,(ft/sec)$	2.758×10^2	-6.188×10^3	-3.515×10^4
$q\,(rad/sec)$	0	-1.017×10^5	-1.122×10^7
$\dot{w}\,(ft/sec^2)$	0	1.308×10^2	-3.826×10^3
$\delta_e\,(rad)$	-3.717	-3.551×10^5	-3.839×10^7

Longitudinal numerical model I

The aircraft configuration means the aircraft mass $m = 19792$ *slug* and the CoG position to be $(0.25, 0, 0)\ \bar{c}$, while the configuration point assumes the altitude $h = 40000$ *ft* for the Mach number $\mathrm{M} = 0.8$ (the steady-state speed is 774 *fps*).

The flight condition is therefore straight and level flight at fixed altitude.

Then, the full longitudinal state and control matrices are as follows

$$A_{LON} = \begin{bmatrix} -0.0069 & 0.0139 & 0 & -32.2000 \\ -0.0904 & -0.3147 & 773.9766 & 0 \\ 0.0001 & -0.0010 & -0.4284 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix} \text{ and}$$

$$B_{LON} = \begin{bmatrix} -0.0002 & -18.0610 & -1.1577 & 0 \end{bmatrix}^{\top}.$$

Longitudinal numerical model II

The eigenvalues of the system matrix A_{LON} provide to be in complex-conjugate pairs

$$\Lambda(A_{LON}) = \{-0.3717 \pm 0.8873\jmath, -0.0033 \pm 0.0672\jmath\} ,$$

thus revealing the natural behaviors, namely the short-period and phugoid modes, with the corresponding eigenvectors

$$\overline{v}_{SP_{1,2}} = [0.0211 \pm 0.0166\jmath, 0.9996, -0.0001 \pm 0.0011\jmath, 0.0011 \pm 0.0004\jmath]^T ,$$
$$\overline{v}_{PH_{1,2}} = [-0.9983, -0.0573 \pm 0.0097\jmath, -0.0001, 0.0001 \pm 0.0021\jmath]^T .$$

See the characteristics of the two modes in Tab., where T (sec) is the period (time constant) and T_h (sec) is the time-to-half, while ζ and ω (rad/s) represent the damping coefficient and the undamped natural frequency, respectively.

Mode	T	T_h	ω	ζ
Short-period	7.08	1.86	0.9621	0.3864
Phugoid	93.47	210.00	0.0673	0.0489

Short-period numerical model I

This far, extract the numerical form of the short-period state and control matrices as follows

$$A_{SP} = \begin{bmatrix} -0.3127 & 774.0000 \\ -0.0010 & -0.4284 \end{bmatrix} \text{ and}$$

$$B_{SP} = \begin{bmatrix} -17.9416 \\ -1.1577 \end{bmatrix},$$

from where the eigenvalues of the short-period system matrix A_{SP} in the set

$$\Lambda(A_{SP}) = \{-0.3706 \pm 0.8779\jmath\}$$

provide the following dynamical parameters, precisely damping coefficient $\zeta_{SP} = 0.387$ and natural frequency $\omega_{SP} = 0.962$ rad/sec.

The corresponding eigenvectors are

$$\overline{v}_{SP_{1,2}} = [1.000, -0.0001 \pm 0.0011\jmath]^T ,$$

from where the initial condition is

$$x_{SP_0} = [1.000, -0.0001]^T .$$

Phugoid numerical model I

Moreover, extract the numerical form of the phugoid state and control matrices as follows

$$A_{PH} = \begin{bmatrix} -0.0069 & -32.2000 \\ 0.0001 & 0.0000 \end{bmatrix} \text{ and}$$

$$B_{PH} = \begin{bmatrix} -15.2196 \\ 0.4180 \end{bmatrix},$$

from where the eigenvalues of the phugoid system matrix A_{PH} in the set

$$\Lambda(A_{PH}) = \{-0.0034 \pm 0.0566\jmath\}$$

provide the following dynamical parameters, precisely damping coefficient $\zeta_{PH} = 0.0489$ and natural frequency $\omega_{PH} = 0.0673$ rad/sec.

The corresponding eigenvectors are

$$\overline{v}_{PH_{1,2}} = [1.000, -0.0001 \pm 0.0018\jmath]^T,$$

from where the initial condition keeps the same values as in the short-period case.

Questions? vs. Answers!

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